

Installing OpenClaw Locally

Your complete step-by-step reference for Mac, Linux, and Windows

This guide is a companion to the *Installing OpenClaw Locally* lesson. Use it as a reference while you follow along with the video, or come back to it later when you need to check a command, fix an issue, or set up a free model. Everything you need to install and run OpenClaw on your own computer is in here.

You don't need a paid AI subscription to follow this guide. The video lesson uses a ChatGPT subscription as the primary example because that's what I personally recommend, but this guide includes complete step-by-step instructions for setting up free alternatives like MiniMax and Google Gemini Flash. So no matter your budget, you have a clear path forward.

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1. Prerequisites

Before running the installer, make sure you have these two things ready on your machine.

Node.js version 22 or higher

OpenClaw requires Node.js 22+. The good news: the installation one-liner will install Node.js automatically if you don't have it, so most people don't need to do anything manual here. To check if you already have Node.js and what version:

```
node --version
```

If it returns a version of 22 or higher, you're good. If it returns an older version or says "command not found," don't worry — the installer handles it.

Git

Git is a version control tool used by OpenClaw behind the scenes during installation. On Mac and Linux, Git is usually already installed, or it gets installed automatically the first time you try to use it. On Windows, Git is **not** included in the one-liner, so you need to install it manually before running the installation.

Windows users: Download Git from git-scm.com/download/win, run the installer, and click through with all the default options. It takes about a minute.

That's it for prerequisites. No Docker, no Python, no complicated setup. Just Node.js (auto-installed) and Git (Windows only). The OpenClaw installer handles everything else.

2. Installing OpenClaw

Head to openclaw.ai and scroll down to the Quick Start section. You'll see two version options: **stable** and **beta**. Always pick stable unless you specifically want to test unreleased features. Copy the one-liner command.

Opening your terminal

On Mac:

Press **Command + Space** to open Spotlight, type *terminal*, and hit Enter. Paste the one-liner and press Enter.

On Windows:

Open **PowerShell as Administrator**. Right-click the PowerShell icon and select "Run as administrator." This avoids several small permissions issues during setup. Paste the one-liner and press Enter.

On Linux:

Open your terminal the way you normally would. Paste the one-liner and press Enter.

The installation can take anywhere from one to a few minutes. The installer will check for dependencies, download OpenClaw, install Node.js if missing, and set everything up. Don't close the terminal or touch anything — just let it finish.

When the installation finishes, the onboarding wizard starts automatically. If for some reason it doesn't, you can start it manually with:

```
openclaw onboard
```

3. Windows Special Notes

Windows is the trickiest platform for running OpenClaw because of how permissions and services work. Here are the things to know.

Always run PowerShell as Administrator

Right-click PowerShell and select "Run as administrator" before starting the installation. This avoids permission issues with the gateway service later on.

WSL2 — an alternative that often runs smoother

WSL2 stands for Windows Subsystem for Linux. It's a way to run a full Linux environment inside Windows, and OpenClaw plays more smoothly inside it than on native Windows. I'd recommend trying native Windows first because it's simpler, but if you run into weird issues or errors, WSL2 is often the fix.

To install WSL2, open PowerShell as Administrator and run:

```
wsl --install
```

Reboot your machine when prompted, and WSL2 will be ready. Microsoft has a full guide at learn.microsoft.com/en-us/windows/wsl/install.

Dashboard won't load? Here's the fix

If you finish the onboarding and the gateway dashboard shows a blank page, an error, or just doesn't load, it's almost always a permissions issue with how the gateway service got installed. The fix is simple:

1. Close your current PowerShell window
2. Open a new PowerShell window **as Administrator**
3. Run the following command:

```
openclaw gateway install
```

This reinstalls the gateway service with the correct permissions. Refresh the dashboard in your browser and it should load properly.

This fix only applies to Windows. Mac and Linux users don't usually run into this issue at all.

4. Security Warning

Please read this before continuing. When OpenClaw runs on your own computer, it can do anything you can do on that machine. It can open your files, run commands, and take real actions. If a bad prompt or a prompt injection tricks the agent into doing something it shouldn't, the damage happens on your own machine. Make sure you're okay with that before you carry on.

Why bigger models are actually safer

This sounds backwards at first, but a smarter AI model actually makes your setup safer, not more dangerous. Here's why. Smaller and weaker models tend to follow whatever instructions they're given without questioning them, which makes them easy to trick. Bigger models like GPT 5.4 or Claude are much better at noticing when something looks off and saying no. So if security matters to you, picking a strong model is one of the best things you can do.

Good habits to start with

- Only give your agent the permissions it actually needs
- Get into the habit of checking what your agent is about to do before approving it
- Set limits so a task can't loop forever if something goes wrong
- Don't connect every service you own on day one
- Start small, test things, and add more later

5. Onboarding Wizard Walkthrough

Here's every option the wizard will ask about, with what to pick and why.

Quick Start vs Manual Configuration

Pick **Quick Start**. It uses sensible defaults for everything and is what I recommend for pretty much everyone getting started. Manual gives you more control but also more ways to get lost.

AI Model Selection

This is where you tell OpenClaw which AI brain to use. You'll see options like Anthropic, OpenAI, Google, OpenRouter, Moonshot, and more. My main recommendation is **OpenAI Codex** (through a ChatGPT subscription via OAuth) — see Section 6 for the full walkthrough. If you want a free option, jump to Section 7 (MiniMax) or Section 8 (Gemini Flash).

Workspace Directory

This is where OpenClaw stores all your files: memory, configuration, skills, and workspace. **Accept the default** — it's a hidden folder called *.openclaw* inside your home directory. On Mac, that's */Users/yourname/.openclaw*.

Package Manager (NPM, PNPM, BUN)

You'll be asked which package manager to use for installing dependencies. BUN is faster, but I recommend **NPM** for simplicity and compatibility — it works reliably across all systems.

Gateway Port

The gateway is the background service that handles all communication. **Accept the default port (18789)** unless you know that port is being used by something else on your machine.

Gateway Bind

This determines how the gateway accepts incoming connections. For a local installation on your own machine, **pick localhost**. It's the most secure option because it only accepts connections from your own computer.

Gateway Token

This is the password that protects your dashboard. **Leave it blank** so the wizard generates a strong random one for you. Auto-generated is way more secure than anything you'd make up yourself.

Messaging Channel

The wizard shows you a long list of chat platforms — Telegram, WhatsApp, Discord, Slack, iMessage, Signal, Google Chat, and a bunch more. **Go with Telegram** for this course. It's the quickest one to set up, it works smoothly across your phone, desktop and tablet, and it's what every other lesson in the course uses. The wizard will ask for a bot token, which you can either provide right away or skip and come back to later.

Skills

Skills are how your agent does real things in the world. The wizard shows you a long list, but don't try to enable everything at once. Every skill you turn on adds more permissions, and it's much safer to start small. **Just pick Clawhub for now** and nothing else. Clawhub is a built-in marketplace, so once your setup is running you can add more skills whenever you actually need them.

Hooks

Hooks are little automations that run at certain moments in your agent's day, like when it starts up, shuts down, or finishes a task. **Go with boot.md** from the list. This one makes sure your agent loads its identity and context every time the gateway starts up, so your agent feels consistent between sessions instead of feeling like a brand new thing every time.

6. Primary Setup: ChatGPT Subscription (OAuth)

This is the setup I personally use and recommend for most people. If you already pay for ChatGPT Plus (\$20/month), you can connect it directly to OpenClaw and use GPT 5.4 through OAuth at no extra per-token cost.

Why this is my top pick: OpenAI officially supports this method, the setup is clean and fast, and you get access to GPT 5.4 as part of your flat monthly fee. No surprise bills, no worrying about token counts.

Step-by-step OAuth flow

Step 1: When the wizard shows you the list of models, scroll down until you find **OpenAI Codex** and pick it.

Step 2: The wizard will either open a browser window for you automatically, or print a link in the terminal that you need to copy. If a browser opens by itself, skip ahead to Step 4. Otherwise, copy the link from the terminal.

Step 3: Paste the link into your browser. You'll see the ChatGPT login page. Sign in with the same account you use at chatgpt.com.

Step 4: Once you're logged in, you'll see a screen asking if you want to give OpenClaw permission to use your ChatGPT subscription. Click **Authorize** or **Continue** to approve it.

Step 5: What happens next depends on your browser. The smooth path is that you'll see a nice *Authentication successful* page. If that's what you see, you're done with the browser part and can close the tab.

Step 5 alternative: Some browsers will instead try to redirect you to a localhost address and show you an error page like *this site can't be reached*. Don't worry if that happens — it's normal and the important info is still in the URL itself. Click on the address bar, select the whole URL, and copy it.

Step 6: If you went through the alternative path in Step 5, go back to your terminal where the wizard is waiting, paste the URL in, and hit Enter. The wizard will pull the authorization code out of the URL on its own.

Step 7: Either way, the terminal should now confirm that OpenAI Codex is set up and ready to go. Your ChatGPT subscription is now powering OpenClaw.

Tip: If the OAuth flow gets stuck or times out for any reason, you can always start the whole onboarding over by running *openclaw onboard* again. And don't be put off by the localhost error page if you see it — it shows up for some browsers and not others, and either way you end up with the same working connection.

7. Free Setup Option A: MiniMax (via OpenRouter)

If you don't want to pay for anything, MiniMax through OpenRouter is one of the best free options available. OpenRouter is a service that gives you access to dozens of AI models — including free ones like MiniMax — all through a single API key.

What you'll get: Access to MiniMax M2.5 completely free. Not quite as sharp as GPT 5.4 or Claude, but genuinely good enough for learning, experimenting, and following along with this course.

Step-by-step MiniMax setup

Step 1: Create a free OpenRouter account

Go to openrouter.ai and sign up. You can use Google or GitHub to sign in quickly. No credit card required for free models.

Step 2: Generate an API key

Once logged in, click your profile icon in the top right and select **Keys**. Click **Create Key**, give it a name like "OpenClaw", and click Create. **Copy the key immediately** — OpenRouter only shows it once.

Step 3: Select OpenRouter in the OpenClaw wizard

When the onboarding wizard asks you to choose a model, scroll to **OpenRouter** and select it. The wizard will ask you to paste your API key. Paste the key you copied from OpenRouter and press Enter.

Step 4: Set MiniMax as your model

The wizard will ask which OpenRouter model to use. Enter the model identifier:

```
minimax/minimax-m2.5
```

Press Enter and you're done. Your OpenClaw is now powered by MiniMax through OpenRouter, completely free.

Tip: The same OpenRouter API key gives you access to other free models like NVIDIA Nemotron 3 Super. You can switch between them anytime by changing the model identifier in your OpenClaw config or by telling your agent directly.

8. Free Setup Option B: Gemini Flash (Google)

Google Gemini Flash has one of the most generous free tiers in the industry — 1,500 requests per day and 1 million tokens per minute. For most people learning OpenClaw, that's way more than you'll ever need.

What you'll get: Access to Google Gemini Flash with a massive free tier. Data on the free plan may be used by Google for training, so keep that in mind if you're doing anything sensitive.

Step-by-step Gemini Flash setup

Step 1: Get a Gemini API key

Go to aistudio.google.com and sign in with your Google account. Once logged in, click **Get API key** in the left sidebar, then **Create API key**. Select a project or create a new one, and your key will be generated.

Step 2: Copy your API key

Click the copy icon next to your key to copy it to your clipboard. Store it somewhere safe — treat it like a password.

Step 3: Select Google in the OpenClaw wizard

When the onboarding wizard asks you to choose a model, scroll to **Google** and select it. The wizard will ask for your Gemini API key. Paste the key you copied and press Enter.

Step 4: Choose Gemini Flash as your model

The wizard will ask which Gemini model to use. Pick a Flash variant — it's the fastest and has the best free tier. The exact model name may be something like *gemini-2.5-flash* or similar, depending on what's available when you set this up.

That's it. Your OpenClaw is now powered by Gemini Flash completely free.

Free tier limits: 1,500 requests per day and 1 million tokens per minute. If you hit these limits, your agent will pause until the quota resets. For casual use, you'll almost certainly never hit them.

9. Dashboard Access & First Test

Once the onboarding wizard finishes, it prints a link to your gateway dashboard with a token attached. That first link is what you use to log in initially. After that, you can access the dashboard directly using the URL below.

Your dashboard URL

If you used the quick start setup like we did in this lesson, your dashboard URL will be exactly the same as everyone else's, because we all picked the same default port. Here it is:

```
http://127.0.0.1:18789
```

Bookmark this URL in your browser so it's one click away whenever you want to check on your agent. The dashboard will ask you for your gateway token the first time you log in, and then it will remember you for future sessions.

Where to find your gateway token: The easiest way is to just ask your agent. Open the dashboard chat (or your Telegram if it's already connected) and type something like *what is my gateway token* and the agent will grab it for you in seconds. No need to dig through any files or run any commands. This is a great example of the golden rule of OpenClaw — tell your agent what you want, not how to do it.

Why this only works on your own machine: During onboarding we set the gateway bind to *localhost*, which means the dashboard only accepts connections coming from the same computer where OpenClaw is installed. This is a deliberate security choice and you should keep it that way for local installations. If you ever need remote access, we cover safer options like Tailscale later in the course.

What the dashboard is for

The dashboard is your control center for OpenClaw. You can chat directly with your agent, check logs, see what's happening behind the scenes, manage skills, view session history, track usage, and configure pretty much everything about your setup. That said, the chat inside the dashboard is mainly for debugging and quick testing. **You don't need to use it for day-to-day conversations with your agent.** Once Telegram is connected, that's where most of your real interactions are going to happen.

We have a dedicated lesson later in the course that walks through the entire dashboard interface in detail, so don't feel like you need to explore every section right now.

Quick test to confirm everything works

In the dashboard chat, type:

```
hello
```

Press Enter. If everything is set up correctly, your agent will respond within a few seconds. That confirms OpenClaw is installed, connected to your AI model, and ready to work.

If nothing happens

If your agent doesn't respond or the dashboard is empty:

- On Windows: try the dashboard fix from Section 3
- Check your API key or OAuth connection is valid
- Check your internet connection
- Restart the gateway with *openclaw gateway restart*
- Check the logs with *openclaw gateway logs*

10. Local AI Models Warning

OpenClaw lets you run AI models on your own hardware through a tool called **Ollama**. The appeal is obvious — zero API costs, complete privacy, and everything stays on your machine. For some use cases this is a perfect fit, but there are some real limitations you should understand before going down this path.

Here's the part most tutorials skip over.

OpenClaw lets you run AI models on your own hardware through a tool called **Ollama**. The idea sounds great — no API costs, full privacy, and everything stays on your machine. For some uses this is a really good fit, but there are some real downsides you should know about before going this way.

Here's the part most tutorials don't tell you.

The truth about running local models with OpenClaw: Lighter tasks like answering questions, having a chat, or writing some code work fine on local setups. But the moment you ask your agent to do something more agentic — browsing websites, running multi-step workflows, or chaining several tools together — the difference between local and cloud models becomes clear very fast. OpenClaw needs the AI model to be both fast and smart at the same time, and most regular computers just can't run a model that's good enough at both.

When local models make sense

- Privacy is your top priority and nothing can leave your machine
- You have a powerful computer (think \$1,000+ with lots of RAM)
- You mostly use OpenClaw for chat, Q&A, or simple coding
- You're curious about local AI and want to learn how it works

When cloud models are the better pick

- You want OpenClaw to actually work like a real agent
- Your tasks involve browsing, automation, and chaining tools
- You care about how fast and snappy the agent feels
- Your computer is average rather than high-end

A middle ground

If your main worry is keeping costs low but you still want OpenClaw to work well, try a free cloud model first. MiniMax via OpenRouter (Section 7) or Gemini Flash (Section 8) both cost nothing and give you much better results than what you'd get from most local models running on a regular computer.

11. Useful Commands Reference

Keep this section bookmarked. These are the commands you'll use most often.

Command	What it does
<code>openclaw onboard</code>	Start the onboarding wizard from scratch
<code>openclaw gateway start</code>	Start the gateway service
<code>openclaw gateway stop</code>	Stop the gateway service
<code>openclaw gateway restart</code>	Restart the gateway service
<code>openclaw gateway status</code>	Check if the gateway is running
<code>openclaw gateway logs</code>	View gateway logs (useful for debugging)
<code>openclaw gateway install</code>	Reinstall gateway service (Windows fix)
<code>openclaw --version</code>	Show your installed OpenClaw version
<code>openclaw --help</code>	Show all available commands
<code>node --version</code>	Check your Node.js version

Tip: If you forget a command, type `openclaw --help` or just ask your agent directly. Remember the mindset: tell it what you want, not how to do it. "Restart yourself" works just as well as any command.

12. Troubleshooting

Common issues and how to fix them.

Issue: Dashboard is blank (Windows)

Permissions issue with the gateway service. Open PowerShell as Administrator and run:

```
openclaw gateway install
```

Issue: Installation fails with Node.js error

Your Node.js version is older than 22, or the installer can't auto-install it. Manually install Node.js 22+ from nodejs.org and try the one-liner again.

Issue: Git not found (Windows)

You skipped the Git prerequisite. Install Git from git-scm.com/download/win and re-run the installation.

Issue: OAuth flow redirects to broken page

This is not actually a bug. Some browsers handle the OAuth callback by trying to load a localhost address that doesn't exist, which shows up as a scary-looking error message. The fix is simple — grab the full URL from your browser's address bar (the info you need is hidden inside the URL) and paste it into the terminal where the wizard is still waiting.

Issue: Agent doesn't respond in dashboard

- Check your API key or OAuth connection
- Run *openclaw gateway status* to confirm it's running
- Run *openclaw gateway logs* to see what's happening
- Try *openclaw gateway restart*
- Check you have credits/quota on your chosen AI provider

Issue: Port 18789 is already in use

Another program is using the default gateway port. During onboarding, pick a different port number (anything above 1024 that isn't used by other software — try 18790 or 19000).

Issue: Free model feels too slow or weak

Free models have rate limits and lower capability than paid ones. If you hit the ceiling of what's possible with free, consider upgrading to the ChatGPT subscription setup (Section 6) for much better results.

When all else fails

Check the official documentation at docs.openclaw.ai or ask your agent directly — OpenClaw can often diagnose and fix its own problems if you tell it what you're seeing.

13. What's Next

OpenClaw is now installed, connected to an AI model, and ready to work. From this point on, the rest of the course works exactly the same whether you're on a VPS or running locally. Here's what's coming next.

Immediate next steps in the course

- Connecting Telegram properly (dedicated lesson with BotFather walkthrough)
- Understanding and hardening your agent's security
- Setting up API keys and smart routing for cost control
- Connecting Google Workspace for email, calendar, and files
- Adding skills and automation with cron jobs

The golden mindset rule

Tell your agent what you want, not how to do it. When something needs to change in your setup, don't try to dig into config files or memorize commands. Just open a chat with your agent and tell it what you're trying to do — the agent will figure out the technical steps for you. We saw a great example of this during the installation lesson. When Telegram needed to be paired, we could have switched to a terminal and typed out the `openclaw pairing approve` command ourselves, but asking the agent to handle it took two seconds and worked just as well. Use this same approach anywhere you'd be tempted to mess with files or commands directly.

Quick reference links

- Your gateway dashboard: <http://127.0.0.1:18789>
- Official OpenClaw docs: docs.openclaw.ai
- Alternative install methods (Docker, Nix, Ansible, NPM, BUN): docs.openclaw.ai
- OpenClaw showcase: openclaw.ai
- OpenRouter (free models): openrouter.ai
- Google AI Studio (Gemini): aistudio.google.com
- Telegram: telegram.org
- BotFather direct link: t.me/BotFather
- OpenClaw AI Models Guide PDF (also attached to this course)
- Telegram Setup Mini Guide PDF (also attached to this lesson)

OpenClaw Masterclass — Installing OpenClaw Locally Guide

Created for students of the OpenClaw Masterclass on Udemy. Commands and procedures may change as OpenClaw evolves — always check the official docs at docs.openclaw.ai for the latest information.